

Rescuing polynomials: a differential topology perspective

Presentation - CUMC 2025

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24th June 2025

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Smooth maps

Definition (Homeomorphism, Diffeomorphism)

Let $X \subseteq \mathbb{R}^k$, $Y \subseteq \mathbb{R}^l$, $f : X \rightarrow Y$ a map.

- f is **smooth** if all its partial derivatives exist and are continuous (i.e., of class $\mathcal{C}^\infty$).
- f is a **homeomorphism** if it is a continuous bijection, whose inverse function is also continuous (i.e. f^{-1} continuous).
- f is a **diffeomorphism** if it is a smooth homeomorphism, whose inverse function is also smooth (i.e. f^{-1} smooth).

Manifolds : formal definition

Definition (Smooth manifold)

Let $M \subseteq \mathbb{R}^k$. M is a **smooth manifold** of dimension m if it is locally diffeomorphic at every point to an open subset of $\mathbb{R}^m$.

More rigorously, there exists a family of diffeomorphisms $(\varphi_i)_i$ such that each $\varphi_i : U_i \rightarrow V_i \cap M$, where $U_i \subseteq \mathbb{R}^m$ and $V_i \subseteq \mathbb{R}^k$ are open sets.

The **charts** (U_i, φ_i) form an **atlas** $(U_i, \varphi_i)_i$ that covers M with the family $(V_i)_i$.

It is equipped with the subspace topology induced by the Euclidean norm $\|\cdot\|_2$ from $\mathbb{R}^k$ (thus $W_i := V_i \cap M$ is open in M).

Manifolds : intuition

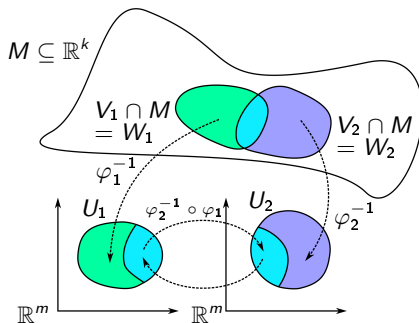


Figure: Intuition behind the construction of an atlas on a surface

Derivatives : a beginning

Manifolds allow us to generalize infinitesimal calculus on much more abstract sets.

Definition (Derivative between open sets)

Let $U \subseteq \mathbb{R}^k$, $V \subseteq \mathbb{R}^l$ be open sets and $f : U \rightarrow V$ a smooth map. The **derivative** (or differential) of f at the point $x \in U$ is the function:

$$d_x f : \mathbb{R}^k \rightarrow \mathbb{R}^l$$
$$h \mapsto \lim_{t \rightarrow 0} \frac{f(x + th) - f(x)}{t}.$$

In fact, $d_x f(h)$ is the directional derivative of f at the point x towards h . Moreover, it is a linear map, for which the jacobian of f evaluated at x is the matrix :

$$d_x f(h) = J_f(x) \cdot h.$$

Tangent spaces

Let us generalize the notion of tangent planes, seen in calculus, to manifolds.

Definition (Tangent space)

Let $M \subseteq \mathbb{R}^k$ be a smooth manifold of dimension m and $x \in M$. Choose a chart (U, φ) such that $\varphi(U)$ an open neighborhood of x and $\varphi(u) = x$. One can consider $\varphi : U \rightarrow \mathbb{R}^k$, in order to obtain that the function $d_u\varphi : \mathbb{R}^m \rightarrow \mathbb{R}^k$ is well defined.

The **tangent space** of M at x is defined as :

$$T_x M = d_u\varphi(\mathbb{R}^m).$$

- $T_x M$ is well defined, meaning that it is independent of the choice of (U, φ) .
- $T_x M$ is a vector space of dimension m , just like M .

Derivatives : general definition

We can now define the derivative of a smooth function whose domain and codomain are smooth manifolds.

Definition (Derivative between manifolds)

Let $M \subseteq \mathbb{R}^k$, $N \subseteq \mathbb{R}^l$ be smooth manifolds, and $f : M \rightarrow N$ be a smooth map. The **derivative** (or differential) of f at the point $x \in M$, where $f(x) = y$, is the linear map:

$$d_x f : T_x M \rightarrow T_y N$$

defined as follows : for U a small neighborhood of x in $\mathbb{R}^k$, take $F : U \rightarrow \mathbb{R}^l$ a smooth extension of $f|_{U \cap M}$. We define $d_x f(h) = d_x F(h)$, $\forall h \in T_x M$.

- $d_x f$ is well defined, meaning that it is independent of the choice of F .
- $d_x f$ is a linear map, by restriction.

Regular points and values

Let $f : M \rightarrow N$ be a smooth map between n -dimensional smooth manifolds.

Definition (Regular value)

$x \in M$ is a **regular point** of f if $d_x f$ is a linear isomorphism.

$y \in N$ is a **regular value** of f if every point $x \in f^{-1}(y)$ is a regular point.

(Note : $f^{-1}(y)$ may be empty.)

If $d_x f$ is not an isomorphism, x is called **critical point**, and $f(x)$ is a **critical value**.

Thus, $y \in N$ is a regular value if and only if $f^{-1}(y)$ contains no critical point.

Theorem

If M is compact, then $\#f^{-1}(y)$ is finite and locally constant, where y ranges through the regular values of f .

The fundamental theorem of algebra

Theorem (Fundamental theorem of algebra)

Each $\mathbb{C}$ -valued nonconstant polynomial $P \in \mathbb{C}[x]$ admits a root in $\mathbb{C}$:

$$\exists x \in \mathbb{C}, P(x) = 0.$$

Proof of FTA (1)

Consider P a nonconstant complex valued polynomial of degree n :

$$P : z \mapsto \sum_{i=0}^n a_i z^i, (a_i)_i \in \mathbb{C}, a_n \neq 0, n \geq 1.$$

Let's identify $\mathbb{C}$ with $\mathbb{R}^2 \times \{0\} \subseteq \mathbb{R}^3$ by $z = a + ib \leftrightarrow (a, b, 0)$.
 $\mathbb{R}^2 \times \{0\}$ is diffeomorphic to $\mathbb{S}^2 \setminus \{(0, 0, 1)\}$ by north stereographic projection or to $\mathbb{S}^2 \setminus \{(0, 0, -1)\}$ by south stereographic projection.

Proof of FTA (2)

More precisely, the north projection is defined as

$$\begin{aligned}\pi_+ : \mathbb{S}^2 \setminus \{(0, 0, 1)\} &\rightarrow \mathbb{R}^2 \times \{0\} \\ (x_1, x_2, x_3) &\mapsto \left(\frac{x_1}{1 - x_3}, \frac{x_2}{1 - x_3}, 0 \right)\end{aligned}$$

with inverse

$$\begin{aligned}\pi_+^{-1} : \mathbb{R}^2 \times \{0\} &\rightarrow \mathbb{S}^2 \setminus \{(0, 0, 1)\} \\ z = (a, b, 0) &\mapsto \left(\frac{2a}{1 + a^2 + b^2}, \frac{2b}{1 + a^2 + b^2}, \frac{-1 + a^2 + b^2}{1 + a^2 + b^2} \right),\end{aligned}$$

π_+ is indeed an diffeomorphism : both maps are smooth.

Proof of FTA (3)

The south projection is defined as

$$\begin{aligned}\pi_- : \mathbb{S}^2 \setminus \{(0, 0, -1)\} &\rightarrow \mathbb{R}^2 \times \{0\} \\ (x_1, x_2, x_3) &\mapsto \left(\frac{x_1}{1+x_3}, \frac{x_2}{1+x_3}, 0 \right)\end{aligned}$$

with inverse

$$\begin{aligned}\pi_-^{-1} : \mathbb{R}^2 \times \{0\} &\rightarrow \mathbb{S}^2 \setminus \{(0, 0, -1)\} \\ z = (a, b, 0) &\mapsto \left(\frac{2a}{1-a^2-b^2}, \frac{2b}{1-a^2-b^2}, \frac{-1-a^2-b^2}{1-a^2-b^2} \right),\end{aligned}$$

π_- is an diffeomorphism for the same reason.

Proof of FTA (4)

Recall that P is defined from $\mathbb{R}^2 \times \{0\}$ to itself. Using projections, we would like to define a function f from $\mathbb{S}^2$ to itself, that "corresponds" to P . Let

$$f : \mathbb{S}^2 \rightarrow \mathbb{S}^2$$
$$(x_1, x_2, x_3) \mapsto \begin{cases} (\pi_+^{-1} \circ P \circ \pi_+)(x_1, x_2, x_3), & \text{si } (x_1, x_2, x_3) \neq (0, 0, 1) \\ (0, 0, 1), & \text{else.} \end{cases}$$

We must prove that f is smooth on all $\mathbb{S}^2$.

Obviously, f is smooth on $\mathbb{S}^2 \setminus \{(0, 0, 1)\}$. It remains to show that it is also smooth in a neighborhood $(0, 0, 1)$.

Proof of FTA (5)

Let's introduce

$$Q : \mathbb{R}^2 \times \{0\} \rightarrow \mathbb{R}^2 \times \{0\}$$
$$z \mapsto (\pi_- \circ f \circ \pi_-^{-1})(z),$$

so that, for $z \neq 0$,

$$Q(z) = \pi_- \circ (\pi_+^{-1} \circ P \circ \pi_+) \circ \pi_-^{-1}(z) = (\pi_- \circ \pi_+^{-1}) \circ P \circ (\pi_+ \circ \pi_-^{-1})(z).$$

Using previous formulas, we obtain that :

$$\pi_+ \circ \pi_-^{-1}(z) = \pi_+ \circ \pi_-^{-1}(a, b, 0) = \left(\frac{a}{a^2 + b^2}, \frac{b}{a^2 + b^2}, 0 \right) = \frac{z}{|z|^2} = \frac{1}{\bar{z}}$$

$$\pi_- \circ \pi_+^{-1}(z) = (\pi_+ \circ \pi_-^{-1})^{-1}(z) = \frac{1}{\bar{z}}$$

since $\frac{1}{\frac{1}{\bar{z}}} = z$.

Proof of FTA (6)

Thus, if $z \neq 0$,

$$\begin{aligned} Q(z) &= (\pi_- \circ \pi_+^{-1}) \circ P \circ (\pi_+ \circ \pi_-^{-1})(z) = \frac{1}{P\left(\frac{1}{\bar{z}}\right)} \\ &= \frac{1}{\sum_{i=0}^n a_i \left(\frac{1}{\bar{z}}\right)^i} = \frac{1}{\sum_{i=0}^n \bar{a}_i \cdot \frac{1}{z^i}} = \frac{z^n}{\bar{a}_0 z^n + \bar{a}_1 z^{n-1} + \dots + \bar{a}_n} \end{aligned}$$

By assumption, $a_n \neq 0 \Rightarrow \bar{a}_n \neq 0$ so that $\lim_{z \rightarrow 0} Q(z) = 0 = Q(0)$. Thus, Q is continuous at 0, and Q being a quotient of polynomials, it must be smooth in a neighborhood of 0.

However, $Q = \pi_- \circ f \circ \pi_-^{-1} \Rightarrow f = \pi_-^{-1} \circ Q \circ \pi_-$ and $\pi_-(0, 0, 1) = 0$. Since π_-^{-1} is smooth around $0 = (0, 0, 0)$, f is necessarily smooth around $(0, 0, 1)$. Thus, f is smooth entirely on $\mathbb{S}^2$.

Proof of FTA (7)

- Recall that $f = \pi_+^{-1} \circ P \circ \pi_+$ on $\mathbb{S}^2 \setminus \{(0,0,1)\}$, where π_+ is a diffeomorphism.

Thus, the critical points of f correspond to the critical points of P , being the zeros of P' (and eventually $(0,0,1)$). However, $P \neq \text{cst} \Rightarrow P' \neq 0$ so that the set of zeros of P' is finite. Therefore, the set of critical points of f is finite.

- Denote $\mathcal{C}$ the set of the critical points of f (with $f(\mathcal{C})$ the set of the critical values of f which is finite), and $\mathcal{R}$ the set of its regular values. Then, $\mathcal{R} = \mathbb{S}^2 \setminus f(\mathcal{C})$.

Proof of FTA (8)

- $\#f^{-1}$ is locally constant on $\mathcal{R}$. Since $\mathbb{S}^2$ is connected and $f(\mathcal{C})$ is finite, $\mathcal{R} = \mathbb{S}^2 \setminus f(\mathcal{C})$ is also connected. Thus, $\#f^{-1}$ is globally constant on $\mathcal{R}$.
- Each critical value has a preimage, by definition.
- Moreover, $\#f^{-1} \neq 0$ on $\mathcal{R}$. Otherwise, $f(\mathbb{S}^2)$ would be equal to $f(\mathcal{C})$ (finite), but is also connected (as the continuous image of a connected set) : it would either have a cardinality of 0 or 1. But $f(\mathbb{S}^2)$ has at least 2 distinct images, since P is not constant and π_+^{-1} is injective. Thus, each regular value has a preimage.
- Then, $f : \mathbb{S}^2 \rightarrow \mathbb{S}^2$ is surjective.
- In particular, one can affirm that there exists $z_0 \in \mathbb{S}^2$ such that $f(z_0) = (0, 0, -1) \in \mathbb{S}^2$. But $f(z_0) = \pi_+^{-1} \circ P \circ \pi_+(z_0)$ since $f(z_0) \neq (0, 0, 1)$, so that

$$P(\pi_+(z_0)) = \pi_+(f(z_0)) = \pi_+(0, 0, -1) = 0.$$

Bibliography